

Requirements Elicitation Report

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Document	Requirements Elicitation Report
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Table of Contents

1. Executive Summary 4

2. Project Context and Objectives 4

2.1 Product Overview 4

2.2 Scope 4

2.3 Elicitation Objectives 5

3. Elicitation Methodology 5

4. Stakeholder Identification 5

5. Survey Findings 6

5.1 Main Problems 6

5.2 Priority Findings 6

6. Developer and Project Team Discussion 7

6.1 Attendance System 7

6.2 Meeting State and Recovery 7

6.3 Implementation-Aware Considerations 7

7. Stakeholder and Expert Interviews 7

8. Consolidated Functional Requirements 8

8.1 Account and Access 8

8.2 Starting and Joining 8

8.3 Waiting Room, Roles and Access 8

8.4 Audio, Video and Screen Sharing 8

8.5 Translation and Captions 9

8.6 Chat and Messaging 9

8.7 Participation 9

8.8 Whiteboard and Collaborative Presentation 9

8.9 Host Controls 10

8.10 Meeting Tools and Outputs 10

8.11 Attendance and Meeting Records 10

8.12 Personalisation, Apps and Support 10

8.13 Recovery 10

9. Non-Functional Requirements 11

10. Business Rules 12
11. Constraints 12
12. Requirements Traceability 13
13. Conflicts, Concerns and Risks 13
13.1 Conflicts 13
13.2 Concerns 13
13.3 Risks 14
14. Open Questions 14
15. Preliminary Prioritization 14
16. Next Steps 15
17. Conclusion 15
Appendix A – Survey Summary 15
Appendix B – Source Mapping and Interview Contributions 16
Appendix C – Documentation Note 16

1. Executive Summary

This report documents the requirements elicitation activities performed for VoidMeet, a video-meeting platform designed to support communication across language barriers through live captions, translation and optional translated speech. The platform is intended to support web access together with Android and Windows applications.

The elicitation process consolidates three sources: a User Requirements & Experience Survey, a Developer and Project Team Discussion, and interviews with selected stakeholders and technical experts. The resulting requirements cover authentication, meeting creation and joining, waiting rooms, roles and permissions, audio/video, screen sharing, captions and translation, chat, reactions, whiteboard collaboration, recording, transcripts, live streaming, personalization, support, recovery, and attendance reporting.

The updated elicitation also captures expert suggestions for code formatting and annotation during collaborative presentation, whiteboard image export, transcripts, Minutes of Meeting, and an attendance system that records participant join time, leave time and duration of presence.

2. Project Context and Objectives

2.1 Product Overview

VoidMeet is a web-based video meeting application in which participants can communicate through audio and video while receiving speech as captions or translations in their selected languages. The project also provides Android and Windows application access.

2.2 Scope

- Account authentication and sign-out.
- Instant, later and scheduled meetings.
- Joining through links or room codes.
- Pre-join camera, microphone and language selection.
- Waiting rooms and controlled admission.
- Host, admin, editor and member roles.
- Audio/video and screen sharing.
- Live captions, translation and optional translated speech.
- Group chat, direct messages, reactions and raise hand.
- Collaborative whiteboard with editing and late-joiner history.

- Host permissions, mute-all and participant removal.
- Recording, timer, transcripts and live streaming.
- Themes and Android/Windows applications.
- Problem reporting and restart recovery.
- Attendance tracking and post-meeting attendance reporting.
- Minutes of Meeting and meeting outputs.

2.3 Elicitation Objectives

- Identify stakeholder needs and problems with existing meeting platforms.
- Convert expectations into testable functional and non-functional requirements.
- Identify business rules, technical constraints and operational concerns.
- Capture expert recommendations that improve collaboration and meeting management.
- Provide traceability from elicitation sources to requirements.

3. Elicitation Methodology

Source	Purpose	Main contribution
User Requirements & Experience Survey	Collect user experiences, pain points and feature priorities.	Problems with existing platforms; priorities for captions, translation, whiteboard, recording, privacy, usability and network efficiency.
Developer and Project Team Discussion	Identify implementation-aware and operational requirements.	Attendance system, join/leave tracking, meeting state, recovery and technical feasibility.
Stakeholder and Expert Interview	Capture targeted suggestions from experienced stakeholders and technical experts.	Whiteboard/code formatting, annotation, whiteboard export, transcripts and Minutes of Meeting.

The three named interviewees—Dr. Rehmat Ullah Khattak, Engr. Sumayyea Salahuddin, and Syed Talha Ali Shah—were interviewed for project requirements and feedback. Their contributions are treated as stakeholder/expert interview findings.

4. Stakeholder Identification

Ref	Stakeholder	Concern
S1	Meeting host / admin	Meeting control, admission, safety and permissions.
S2	Participant	Understanding others and being understood.

S3	Presenter / editor	Sharing material and collaborative presentation.
S4	Mobile / app user	Simple operation across phone, browser and desktop apps.
S5	Event organiser	Timed sessions, recordings, transcripts and streaming.
S6	Business owner	Low operating cost, trust and product viability.
S7	Support person	Useful reports for troubleshooting.
S8	System administrator	Deployment, stability and secure secrets.

5. Survey Findings

The survey captured experiences with Zoom, Google Meet, Microsoft Teams, Discord, Skype and WhatsApp. The findings were used to identify recurring problems and prioritize features for VoidMeet.

5.1 Main Problems

- Network instability.
- Difficulty sharing only a selected portion of the screen.
- Lack of captions and real-time translation.
- Language barriers.
- Limited drawing and annotation.
- Complicated interfaces.
- Recording limitations.
- Privacy and security concerns.
- High CPU/RAM usage in some meeting applications.

5.2 Priority Findings

Area	Finding	Implication
Captions	9/10 rated importance 4 or 5.	Live captions are a major requirement.
Translation	7/10 rated it 5; all recorded responses rated it 3+.	Translation is central to the product.
Selective sharing	7/10 rated it 4 or 5.	Controlled screen sharing is important.
Whiteboard	8 Yes; 2 Maybe.	Integrated drawing is strongly supported.
Recording	8/10 rated it 4 or 5.	Recording is important.
Privacy	7/10 rated concern 4 or 5.	Consent and access rules are needed.
Simple UI	7/10 rated it 4 or 5.	Usability should be prioritized.
Slower internet	9/10 rated it 4 or 5.	Network-efficient behavior is important.

Themes	6 Yes; 2 Maybe; 2 don't care.	Useful but not core functionality.

6. Developer and Project Team Discussion

The Developer and Project Team Discussion was used to identify project-specific behavior that may not be fully visible from a general user survey.

6.1 Attendance System

A key requirement identified in the Developer and Project Team Discussion is an attendance system. VoidMeet should record when a participant joins a meeting, when the participant leaves, and the total duration of that participant's presence. The host or meeting organizer should be able to review this information as an attendance report.

6.2 Meeting State and Recovery

The discussion reinforces the need to preserve meeting state such as chat, whiteboard content, settings and relevant participant information so that a server restart does not unnecessarily destroy the active meeting state.

6.3 Implementation-Aware Considerations

- LiveKit is the intended media backend.
- Translation is constrained to free services with fallback engines.
- Secrets must remain outside source code.
- Rate limiting is required to prevent abuse and overload.
- Meeting state is stored in server memory with file-based persistence in the current design.

7. Stakeholder and Expert Interviews

Three targeted interviews were conducted with Dr. Rehmat Ullah Khattak, Engr. Sumayyea Salahuddin, and Syed Talha Ali Shah.

Interviewee	Suggestion	Requirement area
Dr. Rehmat Ullah Khattak	Code formatting / code blocks on the whiteboard.	Whiteboard code formatting.
Dr. Rehmat Ullah Khattak	Annotation over shared applications during screen sharing.	Screen-share annotation.
Engr. Sumayyea Salahuddin	Export whiteboard content as an image.	Whiteboard export.
Syed Talha Ali Shah	Transcripts and Minutes of Meeting.	Meeting outputs and documentation.

8. Consolidated Functional Requirements

8.1 Account and Access

Req #	Description
FR-01	Unauthenticated visitors are redirected to a Google sign-in page.
FR-02	Protected pages and APIs require a signed-in Google session.
FR-03	A signed-in user can log out.

8.2 Starting and Joining

Req #	Description
FR-04	Start an instant meeting.
FR-05	Create a meeting link for later use.
FR-06	Schedule a meeting from a calendar.
FR-07	Join using a link or room code.
FR-08	Copy/share a meeting link.
FR-09	Enter display name and test camera/microphone before joining.
FR-10	Choose speaking and caption languages.
FR-11	Join without a microphone or camera.

8.3 Waiting Room, Roles and Access

Req #	Description
FR-12	Approval-only meetings provide a waiting room.
FR-13	Admins can admit or deny waiting participants.
FR-14	Waiting requests can be cancelled and reconnects do not duplicate them.
FR-15	Admins can choose open or trusted access.
FR-16	The first participant becomes admin; another participant is promoted if the last admin leaves.
FR-17	Admins can assign admin, editor or member roles.
FR-18	Admins can remove participants.

8.4 Audio, Video and Screen Sharing

Req #	Description
FR-19	Participants can mute/unmute microphones and turn cameras on/off.
FR-20	Video tiles show names and device status.
FR-21	Front-camera video is presented naturally.
FR-22	Participants can share their screen when allowed.
FR-23	Participants can spotlight a tile in their own view.

FR-24	Admins can spotlight one participant for everyone.
FR-25	Full-screen and picture-in-picture modes are available.
FR-26	A clear control enables meeting audio when the browser blocks automatic playback.
FR-27	Authorized presenters/editors can annotate over shared screen content.

8.5 Translation and Captions

Req #	Description
FR-28	Participants can select supported speaking and caption languages.
FR-29	Speech can be converted to live captions.
FR-30	Speech segments can be translated into each listener's selected caption language.
FR-31	Listeners can enable spoken playback of translated captions.
FR-32	Speaking and caption languages can be changed during a meeting.
FR-33	A translation panel provides language, voice and status controls.

8.6 Chat and Messaging

Req #	Description
FR-34	Participants can send group chat messages.
FR-35	Admins can pin/delete messages; senders can delete their own messages.
FR-36	Participants can see typing indicators.
FR-37	Private messages require recipient acceptance.

8.7 Participation

Req #	Description
FR-38	Participants can send reactions.
FR-39	Participants can raise/lower a hand.
FR-40	The people panel shows roles, device status and raised hands.

8.8 Whiteboard and Collaborative Presentation

Req #	Description
FR-41	Authorized users can draw on a shared whiteboard.
FR-42	Authorized users can move, resize and clear drawings.
FR-43	Late joiners receive current whiteboard history.
FR-44	Authorized users can insert readable formatted code blocks on

	the whiteboard.
FR-45	Authorized users can export the whiteboard as an image.

8.9 Host Controls

Req #	Description
FR-46	Admins can allow/block screen sharing, reactions, microphone, video and chat.
FR-47	Admins can mute all participants.

8.10 Meeting Tools and Outputs

Req #	Description
FR-48	Authorized users can record and download a meeting.
FR-49	An authorized user can start a meeting timer.
FR-50	An authorized user can start a transcript after a consent warning and download it.
FR-51	Transcript output should retain useful speaker and time context where supported.
FR-52	An authorized user can create structured Minutes of Meeting containing discussion points, decisions and action items.
FR-53	An authorized user can configure YouTube Live streaming.

8.11 Attendance and Meeting Records

Req #	Description
FR-54	The system records each participant's join time and leave time.
FR-55	The system calculates each participant's total presence duration.
FR-56	The host or authorized organizer can view an attendance report containing participant name, join time, leave time and presence duration.

8.12 Personalisation, Apps and Support

Req #	Description
FR-57	Users can choose a theme, custom colour or background image.
FR-58	Android and Windows applications can be downloaded.
FR-59	Users can submit problem reports with contact information, message and optional screenshot.

8.13 Recovery

Req #	Description

FR-60	Meeting settings, board, chat and relevant meeting state are persisted and restored after server restart.

9. Non-Functional Requirements

Req #	Area	Description
NFR-01	Capacity	Up to approximately 50 participants per meeting by default.
NFR-02	Caption speed	Translated captions target approximately an 8-second translation timeout.
NFR-03	Security	Authenticated access, validated room IDs and controlled media-token issuance.
NFR-04	Abuse protection	Rate limits and bounded speech input reduce overload and abuse.
NFR-05	Reliability	Translation engines support fallback behavior.
NFR-06	Availability	The app restarts after crashes and restores persisted meeting state.
NFR-07	Usability	Desktop and mobile interfaces provide clear controls.
NFR-08	Compatibility	Current Chrome, Edge, Firefox and Safari are targeted; speech recognition may depend on browser support.
NFR-09	Privacy	Consent warnings and defined retention/access rules apply to meeting outputs.
NFR-10	Localisation	The configured supported language set is available for speech/captions/streaming.
NFR-11	Portability	Web functionality is packaged for Android and Windows.
NFR-12	Maintainability	Automated tests cover key server, admission, translation, joining and video behavior.
NFR-13	Scalability	Media and translation backends are configurable.
NFR-14	Network efficiency	The application should remain usable on slower or unstable connections where technically possible.
NFR-15	Attendance accuracy	Join/leave events should be recorded consistently enough to calculate reliable presence duration.

10. Business Rules

Rule	Description
BR-01	Only users authenticated with Google may use protected VoidMeet functionality.
BR-02	The first participant becomes admin and a room should retain at least one admin.
BR-03	Trusted mode requires admin approval; open mode permits link-based joining.
BR-04	A meeting cannot exceed the configured participant limit; current default is 50.
BR-05	Only admins can change host settings, roles, removal, pinning, global spotlight and mute-all.
BR-06	Admins and editors can use the whiteboard; members can view it.
BR-07	Messages may be deleted by their sender or an admin.
BR-08	Direct messaging opens only after recipient acceptance.
BR-09	Host-disabled features cannot be used by non-admins.
BR-10	Translation must use free engines under the current project constraint.
BR-11	A transcript requires acknowledgement of the consent warning.
BR-12	Problem reports require name, phone, email and message.
BR-13	Live streaming requires appropriate YouTube access and an event title.
BR-14	Attendance reports are restricted to authorized meeting roles.
BR-15	Transcripts, minutes and other meeting outputs are accessible only to authorized users.

11. Constraints

Constraint	Area	Description
CR-01	Media backend	Production media transport uses LiveKit.
CR-02	Translation cost	Only free translation engines are permitted by the project constraint.
CR-03	HTTPS	HTTPS is required for relevant browser media capabilities on supported devices.
CR-04	Google OAuth	Authentication depends on configured Google OAuth credentials.
CR-05	Speech recognition	Speech recognition depends on browser capabilities.
CR-06	Secrets	API keys and credentials must remain outside source code.
CR-07	Translation request size	Translation services may require text

		segmentation.
CR-08	Meeting state	The current design uses server memory with file persistence.
CR-09	Recording upload	Recording uploads are subject to the configured WebM/server conversion limit.
CR-10	YouTube	Live streaming depends on YouTube account access and availability.
CR-11	Email reports	Support email delivery requires SMTP configuration.

12. Requirements Traceability

Source	Finding	Requirement(s)
Survey	Language barriers / captions	FR-28–FR-33; NFR-02; NFR-10
Survey	Network instability	NFR-01; NFR-06; NFR-14
Survey	Drawing/annotation	FR-41–FR-45
Survey	Recording importance	FR-48
Survey	Privacy concerns	NFR-09; BR-11; BR-15
Developer and Project Team Discussion	Attendance tracking	FR-54; FR-55; NFR-15
Developer and Project Team Discussion	Attendance report	FR-56; BR-14
Developer and Project Team Discussion	Recovery	FR-60; NFR-06
Dr. Rehmat Ullah Khattak – Interview	Code formatting	FR-44
Dr. Rehmat Ullah Khattak – Interview	Screen annotation	FR-27
Engr. Sumayyea Salahuddin – Interview	Whiteboard image export	FR-45
Syed Talha Ali Shah – Interview	Transcripts	FR-50; FR-51
Syed Talha Ali Shah – Interview	Minutes of Meeting	FR-52

13. Conflicts, Concerns and Risks

13.1 Conflicts

- Host control versus quick joining.
- Free translation versus translation accuracy.
- Recording/transcription versus participant privacy.
- Useful meeting records versus retention/storage/privacy concerns.

13.2 Concerns

- Translation delay may make captions arrive after the conversation has moved on.
- Weak mobile connections may reduce media quality.

- Browser speech-recognition support varies.
- Attendance accuracy depends on robust join/leave/reconnect handling.
- Screen annotation requires synchronization between presenter and participants.
- The exact balance between manual and automated Minutes of Meeting needs confirmation.

13.3 Risks

Risk	Impact	Mitigation direction
Translation failure	High	Fallback engines and status messages.
Server overload	High	Rate limits, participant limits and bounded inputs.
Privacy violation	High	Consent, role-based access and retention rules.
Attendance event loss	Medium/High	Persist join/leave events and handle reconnects.
Large recordings	Medium	Upload limits and retention policy.
Annotation synchronization	Medium	Controlled updates and persistent state.

14. Open Questions

1. Should the host be able to disable translation for a meeting?
2. Should participants be notified when recording or streaming starts?
3. What meeting size is expected for the first release?
4. Should scheduled meetings send email invitations?
5. Should attendance reports be downloadable as CSV/PDF or only viewable in VoidMeet?
6. How long should attendance records, transcripts, recordings and minutes be retained?
7. Should Minutes of Meeting support manual entry, automated generation, or both?
8. Should screen annotation be available to all presenters or only explicitly authorized users?
9. Which programming languages should be supported by whiteboard code formatting?

15. Preliminary Prioritization

Priority	Requirement areas	Reason
Must Have	Authentication, joining, audio/video, roles, screen sharing, captions/translation, basic chat, host controls, recovery, attendance tracking/reporting	Core meeting operation and major elicitation needs.
Should Have	Whiteboard, reactions, transcript,	Strong value; can be staged after core

	recording, Minutes of Meeting, annotation, whiteboard export	meeting reliability.
Could Have	Themes, picture-in-picture, live streaming, advanced output retrieval	Useful extensions after core functionality is stable.

16. Next Steps

10. Review the consolidated requirements with the project team.
11. Confirm open questions with relevant stakeholders.
12. Define retention and access rules for attendance, transcripts, recordings and Minutes of Meeting.
13. Transfer approved requirements into the final SRS.
14. Map approved requirements to use cases and system components.
15. Define acceptance criteria and test cases for high-priority requirements.
16. Validate technical feasibility of screen annotation, code formatting, whiteboard export and attendance persistence.

17. Conclusion

The elicitation process provides a consolidated view of what VoidMeet should accomplish from the perspectives of users, developers and interviewed stakeholders. The requirements extend beyond basic video calling to address language accessibility, collaborative presentation, host control, meeting records and operational reliability.

The attendance requirement gives meeting hosts and organizers a structured way to understand participation by recording join time, leave time and presence duration and presenting this information in an attendance report. The stakeholder interviews additionally contribute practical capabilities such as code formatting, screen annotation, whiteboard export and Minutes of Meeting.

The consolidated requirements should be reviewed and formally approved before being carried into the final Software Requirements Specification and implementation plan.

Appendix A – Survey Summary

Finding	Summary
Existing platforms	Zoom, Google Meet, Microsoft Teams, Discord, Skype and WhatsApp were reported.
Satisfaction	Responses were mixed, with approximately half satisfied and

	half neutral.
Captions	9/10 rated importance at 4 or 5.
Translation	7/10 rated real-time translation at 5; all recorded responses rated it 3+.
Selective sharing	7/10 rated it 4 or 5.
Whiteboard	8 Yes; 2 Maybe.
Recording	8/10 rated importance at 4 or 5.
Privacy	7/10 rated concern at 4 or 5.
Simple UI	7/10 rated importance at 4 or 5.
Slower internet	9/10 rated importance at 4 or 5.
Themes	6 Yes; 2 Maybe; 2 don't care.

Appendix B – Source Mapping and Interview Contributions

Source	Contribution incorporated
User Requirements & Experience Survey	Existing-platform problems, feature priorities, privacy concerns, usability and network requirements.
Developer and Project Team Discussion	Attendance system; join time, leave time, presence duration; meeting state and recovery.
Dr. Rehmat Ullah Khattak	Code formatting/code blocks and annotation over shared applications.
Engr. Sumayyea Salahuddin	Whiteboard export as an image.
Syed Talha Ali Shah	Transcripts and Minutes of Meeting.

Appendix C – Documentation Note

This report is a requirements-elicitation artifact for the VoidMeet Software Engineering project. Requirements are presented as consolidated project requirements rather than as direct quotations. Where a requirement originated from an interview or project-team discussion, the source is identified in the traceability and interview sections.